

# Cole Pearson

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## EDUCATION

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### University of California San Diego

La Jolla, CA

Ph.D. in Biochemistry and Molecular Biophysics, Expected 2031

### Shippensburg University

Shippensburg, PA

B.S. in Chemistry (ACS Certified), May 2026

Wood Honors College | Minor: Mathematics | GPA: 3.993

## RESEARCH & DISTINGUISHED ACADEMIC EXPERIENCES

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### Computational Tool Development for Kombucha Titration Analysis

May 2025–August 2026

PIs: Drs. Jeb Kegerreis, John Richardson, Department of Chemistry; Dr. Luis Melara, Department of Mathematics; Shippensburg University, Shippensburg, PA

- Developed a new MATLAB-based computational tool to deconvolute weak acid mixtures
- Built novel model of a multi-acid system from charge balance, mass balance, and aqueous electrolyte theories
- Manuscripts in preparation
- Awarded departmental *Beres Research Grant* (2025–2026) and *SURE Research Grant* (2025)

### NSF REU: Sustainable Chemistry and Catalysis

May 2024–July 2024

PI: Dr. Scott Sayres; Arizona State University, School of Molecular Sciences, Tempe, AZ

- Elucidated increasing excited-state lifetimes of bromoalkanes as a function of carbon chain length
- Operated Ti:Sapphire femtosecond laser, Nd:YAG 800 nm laser, time-of-flight mass spectrometer
- Employed *Gaussian* software to perform quantum mechanical energetic calculations

### DOE/ACS Nuclear Chemistry Summer School (NCSS)

June 2025–July 2025

Lead Instructor: Jeff C. Bryan; San Jose State University, San Jose, CA

- **Trish Baisden Student of the Year Award**; graduated with A+ in both lecture and lab courses
- Completed intensive 6-week, 7-credit lab, lecture, and safety training program
- Studied fundamental nuclear science, fuel reprocessing, medicine, security, radiochemistry, and energy

### MaterialZ Winter School

January 2026

Sponsored by ASU & University of AZ; Biosphere 2, Oracle, AZ

- Materials Science & Engineering seminar series featuring industry and academic leaders
- Presented poster: “Development of a Computational Tool for the Analysis of Kombucha Acid–Base Titration Data”

### German History, Culture, and Geography Course Development

May 2023–August 2023

Advisors: Dr. Alison Feeny & Dr. David Wildermuth, Shippensburg University, Shippensburg, PA

- Compiled course materials and assisted in curriculum development; Awarded SURE Grant (2023)
- Selected as one of top two presentations at SU Summer Research Expo (2023)

## AWARDS AND ACHIEVEMENTS

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- **American Chemical Society (ACS) Award**, Shippensburg University (SU) *May 2026*  
Presented to the outstanding senior Chemistry major, based on demonstrated scholarship, leadership, professional growth, and potential for significant contribution to the advancement of Chemistry.
- **ACS Undergraduate Award in Physical Chemistry**, SU *May 2026*  
Awarded for demonstrated excellence in physical chemistry and related fields based on research, coursework, and/or dedication.
- **ACS Undergraduate Award in Inorganic Chemistry**, SU *May 2026*  
Awarded for demonstrated excellence in inorganic chemistry at the undergraduate level based on any combination of research, coursework, and/or motivation/interest/dedication.
- **Outstanding Honors Capstone Award**, Wood Honors College, SU *April 2026*  
Recognizes excellence in an Honors senior's research, creative, or service-learning Capstone Project.
- **Spirit of Honors Award**, Wood Honors College, SU *April 2026*  
Recognizes an Honors senior who has exemplified the Wood Honors College's commitment to scholarship, leadership, and service and has made exceptional contributions to the Wood Honors College during their undergraduate years.
- **SHORE Graduate Fellowship**, UC San Diego *March 2026*
- **First Place, ACS-AGFD Undergraduate Poster Competition**, Atlanta, GA *March 2026*
- **Dr. James A. Beres and Merry Jean Beres Research Award**, SU *2025-2026*  
Competitive Shippensburg University grant awarded annually to one senior student for outstanding academic achievement, exceptional departmental service, and demonstrated potential in proposed research.
- **Trish Baisden Student of the Year Award**, NCSS *July 2025*
- **Dr. Louise J. Lienemann Chemistry Scholarship**, SU *2024-2026*  
"...awarded to a junior chemistry major demonstrating academic excellence... Character, citizenship, and a cumulative GPA of at least 3.5 on a 4.0 scale are factors in selecting the recipient."
- **ACS Standardized Exams (National Percentile Ranking)**  
General Chemistry (97th) | Physical Chemistry (100th) | Organic Chemistry (96th) | Biochemistry (95th)
- **Shippensburg University Library Research Award, First Prize** *April 2023*  
"The Use of Psychedelics in Therapies Targeting Alzheimer's Disease: A Literature Review"

## PRESENTATIONS (\* denotes regional or national conference/meeting)

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- **Minds@Work Symposium**, Shippensburg University, PA *28 April 2026*  
"Development of a Computational Tool for the Analysis of Kombucha Titrations"
- **Wood Honors College Senior Symposium** *21 April 2026*  
"Development of a Computational Tool for Kombucha Titration Analysis"
- **\*Intercollegiate Student Chemists Convention**, Ursinus College, PA *18 April 2026*  
"Development of a Computational Tool for the Analysis of Kombucha Titrations" – **Second Place**, Physical Chemistry Section
- **Invited Speaker, Department of Mathematics**, Shippensburg University, PA *16 April 2026*  
"Development of a Computational Tool for the Analysis of Kombucha Titrations"

- **\*Mathematical Association of America (MAA) EPaDel Section Meeting**, Shippensburg University, PA 11 April 2026  
"Development of a Computational Tool for the Analysis of Kombucha Titrations"
- **\*American Chemical Society (ACS) National Meeting**, Atlanta, GA 22–26 March 2026  
"Development of a computational tool for kombucha titration analysis"  
**First Place**, ACS–AGFD Undergraduate Poster Competition; \$1,000 competitive travel award; ACS–NUCL/DOE travel award
- **Invited Speaker: Chemistry Club**, Shippensburg University, PA 2 February 2026  
"Preparing Applications for REUs, Internships, and Grad School"
- **\*National Collegiate Honors Council 2025**, San Diego, CA 6–9 November 2025  
"Voices of Peace: Thinking Nobel and Acting Local in Shippensburg, PA"
- **Seminar, Department of Chemistry & Biochemistry**, Shippensburg University, PA 10 October 2025  
"NCSS in San José"
- **\*GCURS 2025**, Rice University, Houston, TX 18 October 2025  
"Development of a Computational Tool for Kombucha Titration Analysis"
- **SURE Summer Research Exposition** 22 August 2025  
"Expansion of a Modeling Method to Determine Weak Acid Identities and Quantities in a Complex Mixture such as Kombucha"
- **Seminar, Department of Chemistry & Biochemistry**, Shippensburg University, PA 8 November 2024  
"REU at ASU: Ultrafast Laser Spectroscopy"
- **NSF REU Oral Infographic Presentation**, ASU, Tempe, AZ 21 July 2024  
Oral presentation and Q&A at School of Molecular Sciences
- **Minds@Work Symposium**, Shippensburg University, PA 23 April 2024  
"Chemistry and German Beer"
- **\*PA Undergraduate Research at the Capitol (URC)**, Harrisburg, PA 19 March 2024  
"German Beer History, Culture, and Geography Course Development"
- **SURE Summer Research Exposition** 25 August 2023  
"German Beer History, Culture, and Geography Course Development"

## SKILLS

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- **Analytical techniques:** Femtosecond laser, gamma spectroscopy, alpha, IR, & NMR, UV/Vis spectroscopies; TOF mass spectrometry; gas & liquid chromatography; potentiometry
- **Class IV Laser Operation & Safety Certification:** Conducted alignment, diagnostics, maintenance
- **Radioactive Material Handling & Safety:** Completed 1-credit radiation safety course and 3-credit advanced nuclear chemistry lab; handled radioactive sources and conducted radionuclide syntheses
- **Biochemical skills:** SDS–PAGE, protein quantification assays, dialysis, ultracentrifugation, enzyme kinetics, and buffer preparation
- **General Laboratory Techniques:** Laser maintenance, organic synthesis, distillation, calorimetry, sample preparation, separations chemistry
- **Computational Skills:** MATLAB® modeling and development; Python – data analysis, model-

ing, and workflow; *Gaussian* (software) – performed quantum mechanical density functional theory calculations

- **Presentation and Teaching:** Formal secondary training in theater and dramatic arts applied to >300 hours of tutoring, TA-ing, and presenting
- **German language:** Intermediate proficiency; High school + 12 credits of college coursework

#### PROFESSIONAL EXPERIENCE

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- **Teaching Assistant, NCSS**, *June–July 2026*, San Jose State University, CA
- **Teaching Assistant, Biochemistry**, *August 2025–May 2026*, Shippensburg University
- **Tutor, General & Organic Chemistry**, *August 2024–May 2026*, Shippensburg University
- **Honors Peer Anchor Teaching Assistant**, *August 2024–May 2026*, Shippensburg University  
Leading first-year cohort through academic and professional transition to college; 10 hrs/week of mentorship
- **Package Handler**, *May 2022–March 2024*, UPS, Carlisle, PA  
Worked 25–30 hours/week along full-time studies; first “Employee of the Month” among 500+ employees

#### CAMPUS INVOLVEMENT / COMMUNITY SERVICE

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- **Co-Founder, Partners in Peace, Nobel Peace Center Collaboration**, *January 2025–May 2026*  
Launched “Voices of Peace” community outreach event; principal author of Nobel Peace Prize Nomination
- **Vice President, Chemistry Club**, *May 2025–May 2026*  
Lead campus/community outreach; host fun events, alumni networking, and local school science nights
- **Treasurer, Kappa Mu Epsilon (Math Honor Society)**, *May 2025–May 2026*
- **Peer Mentor, Department of Chemistry & Wood Honors College**, *August 2024–May 2026*
- **Volunteer, ShipVotes** (nonpartisan voter participation group), *March 2024–May 2026*
- **Academic Appeals Committee Member, SU Department of Chemistry**, *August 2024–May 2026*
- **Volunteer Judge, Scholastic Art & Writing Awards**, *November 2024–Present*
- **Staff News Writer & DJ, WSYC 88.7FM Radio**, *September 2023–May 2025*
- **Volunteer, Honors Service Committee**, *August 2022–May 2026*
- **Chair, Orientation Committee, Wood Honors College**, *January 2023–April 2024*